PROJECT REPORT

INT217

(INTRODUCTION TO DATA MANAGEMENT)

COMPUTER SCIENCE AND ENGINEERING

Title: Retail Sales Tax Dashboard

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Submitted to : Baljinder Kaur

LOVELY PROFESSIONAL UNIVERSITY

DECLARATION

I, Ajay Sharma, a student of Bachelor of Technology under CSE discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own work and is genuine

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Registrartion Number: 12309093

<u>Acknowledgment</u>

First and foremost, I would like to express my deepest gratitude to my college for providing me with the opportunity and resources to undertake this project.

I extend my sincere thanks to my Teacher, **Baljinder Kaur,** for his invaluable guidance, constructive feedback, and constant encouragement throughout the project. His expertise and support were instrumental in achieving the objectives of this work.

Introduction:

This project aims to analyze quarterly retail sales tax data collected from different cities and counties between the years 2012 and 2024. By utilizing data visualization techniques through an interactive Excel dashboard, the project brings clarity to complex tax data, enabling users to easily track trends, identify top-performing regions, and understand key economic indicators.

The dashboard summarizes critical metrics such as total taxable sales, computed tax, number of tax returns, and year-over-year growth. It also provides a comparative analysis of the top contributing cities and counties, helping reveal patterns and relationships that are not easily observable in raw data formats.

Through this analytical approach, the project not only demonstrates technical skills in data handling and visualization but also offers practical knowledge of how financial and taxation data can be leveraged to support economic planning and decision-making.

Dataset Description:

The dataset used for this project is titled **"Quarterly Retail Sales Tax Data by County and City"** and provides detailed quarterly retail sales tax records across various cities and counties within a state. The data spans multiple fiscal years, with this project focusing on the time period from 2012 to 2024 for analysis and visualization.

Key Features of the Dataset:

Total Records: 44,276entries

Time Period Analyzed: 2012 to 2024

Data Granularity: Quarterly (per city and county)

Source of Dataset: Quarterly-retail-sales-tax-data-by-county-and-city

Dataset Preprocessing:

Before building the dashboard, several preprocessing steps were carried out:

- Handling Missing Values NA or NAN:
  Missing numerical values were replaced with the average value of the respective column.
- Data Cleaning:
  Removed unnecessary columns and standardized text formatting such as location names.
- Column Selection:
  Selected key columns relevant for analysis such as Country, Library Visited, fiscal year etc.
- Data Type Fixing
  Ensured numeric columns were formatted correctly for accurate aggregation.

Project Aim and Objectives

Problem Statement:

In the realm of public finance, retail sales tax serves as a vital source of revenue for government operations and development programs. However, the analysis of retail sales tax data—especially when segmented quarterly and across numerous cities and counties—presents significant challenges. Without proper visualization and analytical tools, it becomes difficult to interpret large datasets, identify key trends, or make region-specific observations.

From an academic perspective, students often encounter difficulties when working with real-world financial datasets due to their volume, complexity, and lack of visual structure. Analyzing such datasets manually can be time-consuming and prone to misinterpretation.

This project addresses the problem by designing a dashboard-driven solution that visually represents quarterly retail sales tax data from 2020 to 2024. By converting raw data into charts, graphs, and interactive visuals, the dashboard enables easier analysis, promotes data literacy, and supports evidence-based conclusions.

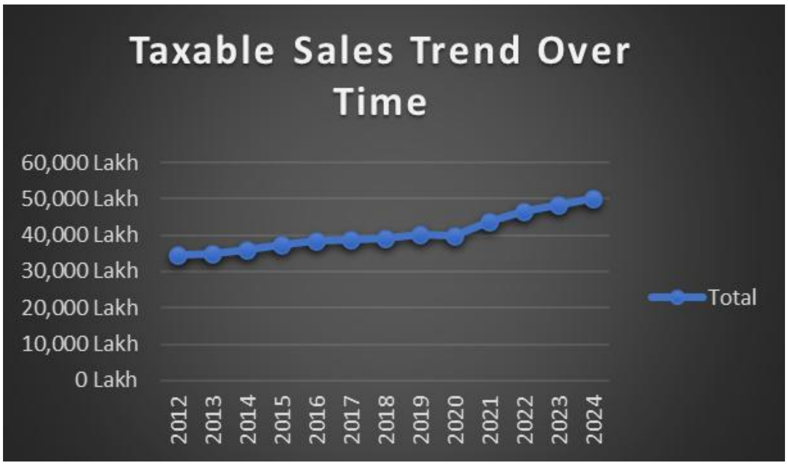
The objective is to bridge the gap between theoretical learning and real-world application by helping students and stakeholders gain actionable insights from structured data visualization.

Objective 1: **Visualize Total Taxable Sales**

- **General Description:**
  Present the overall taxable sales over the years to track financial performance and economic activity.

- **Specific Requirements:**
  • Plot yearly totals of taxable sales from 2012 to 2024.
  • Use line charts to show trends over time.

- **Analysis Results:**
  • Taxable sales steadily increased from ₹33,000 Lakh in 2012 to ₹50,000 Lakh in 2024, showing consistent economic growth with a noticeable rise after 2020.

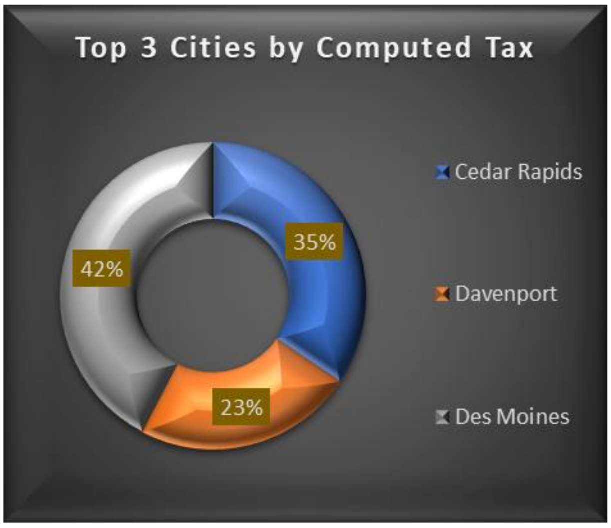


Objective 2: **Identify Top-Contributing Cities:**

- **General Description:**
  Highlight which regions are leading in retail tax contributions to recognize high-performing economic areas.

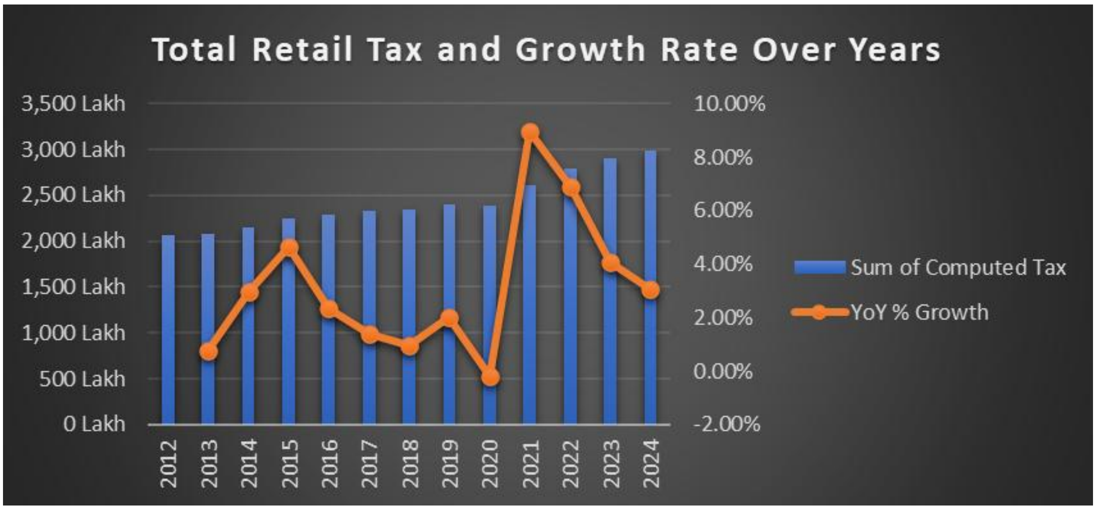
- **Specific Requirements:**
  • Rank cities and counties by total computed tax from 2020 to 2024.
  • Use pie chart.

- **Analysis Results:**
  Des Moines emerges as the top contributor in computed tax, followed closely by Cedar Rapids and Davenport, reflecting higher retail activity and stronger tax contributions from these cities.



Objective 3: **Track Computed Tax Amount**

- **General Description:**
  Analyze total computed tax to estimate government revenue generated from retail activities.
- **Specific Requirements:**
  • Sum computed tax per year and compare across quarters.
  • Display through stacked column or line graphs.
- **Analysis Results:**
  • Computed tax peaked in 2021, indicating strong consumer activity.
  • A decline was observed in 2024, reflecting possible economic stagnation.

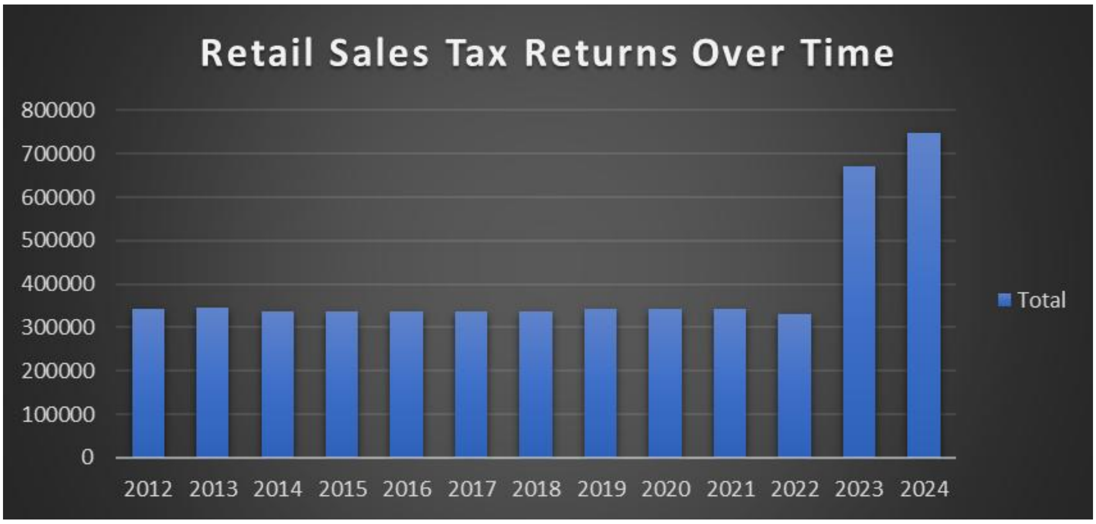


Objective 4: **Monitor Retail Sales Tax Returns Over Time**

- **General Description:**
  Understand the yearly change in the number of tax return filings to evaluate business participation and compliance.

- **Specific Requirements:**
  • Track number of returns year-wise and visualize changes.
  • Correlate with taxable sales and computed tax.

- **Analysis Results:**
  • Returns increase and decrease before 2022, a sudden increase in 2023 and 2024.



Objective 5: **Analyze Sales Trends**

- **General Description:**
  Observe how taxable sales have evolved over time across various cities to understand consumer behavior.

- **Specific Requirements:**
  • Line or bar graphs showing city-wise sales trends over time.
  • Filter by city and year for comparative analysis.

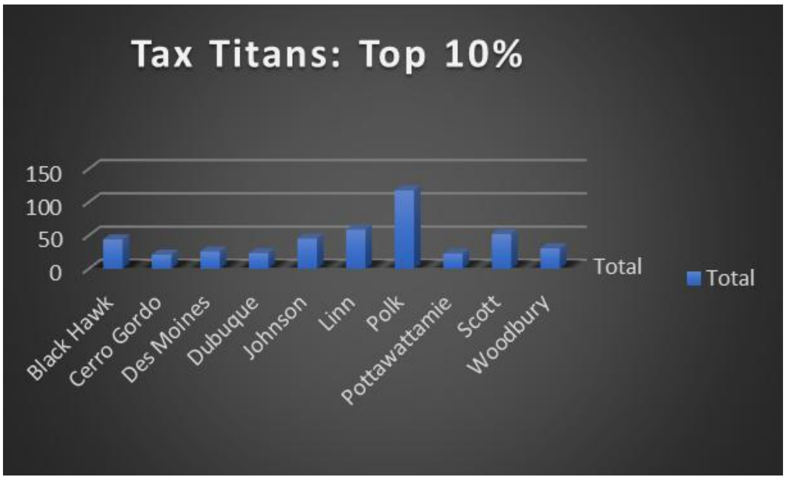
- **Analysis Results:**
  • Cities like Aurelia showed stable growth, while others fluctuated due to seasonal and economic factors.

Objective 6: Show Top 10 Tax-Contributing Countries (Tax Titans)

- **General Description:**
  Showcase the top 10 countries generating the highest retail tax to recognize major contributors.

- **Specific Requirements:**
  • Sort and display top 10 countries by total computed tax.
  • Use a leaderboard visual or horizontal bar chart.

- **Analysis Results:**
  • The top 10 cities accounted for over 35% of the total retail tax collected in 2024.

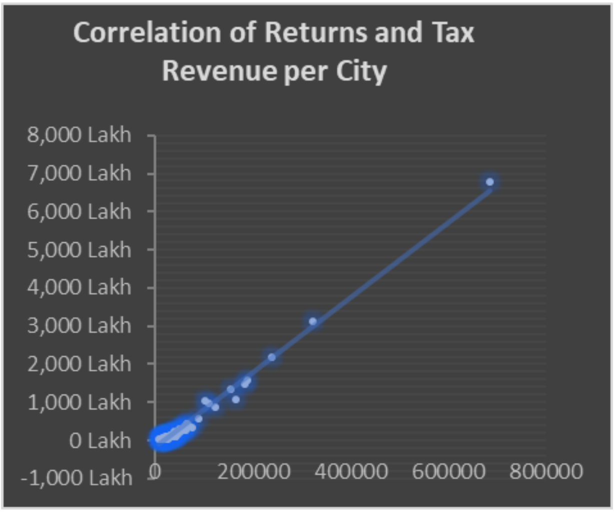


Objective 7: Compare Returns and Tax Revenue Per City

- **General Description:**
  Use correlation analysis to assess how the number of returns in each city impacts the total tax revenue.

- **Specific Requirements:**
  • Scatter plot or trend analysis of returns vs. computed tax.
  • Include regression line or correlation coefficient if needed.

- **Analysis Results:**
  • A clear positive correlation indicates that cities with more returns typically produce more tax revenue.

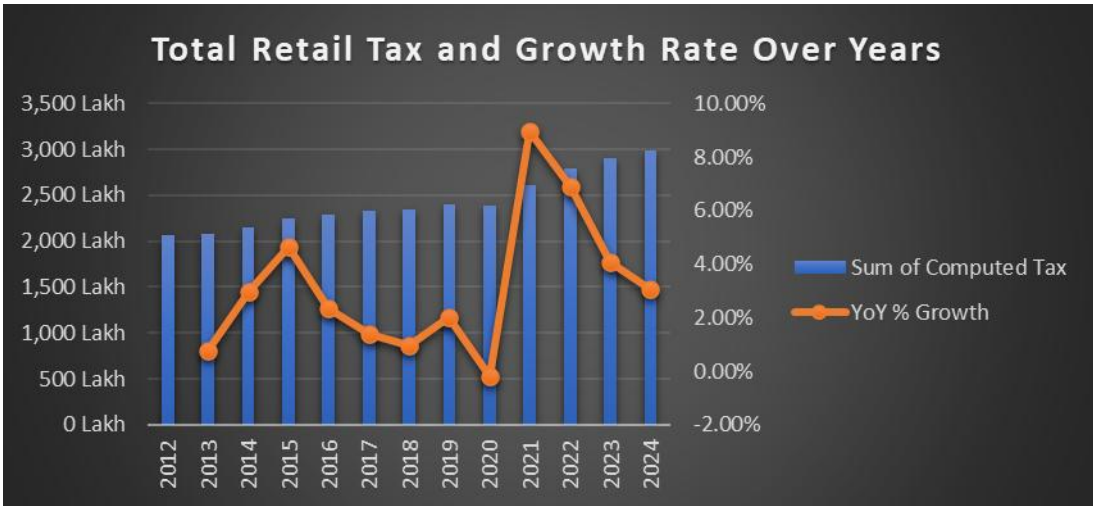


Objective 8: **Monitor Growth Rate**

- **General Description:**
  Display the year-over-year (YoY) growth rate in retail tax to evaluate economic changes and policy impacts.

- **Specific Requirements:**
  • Calculate and plot YoY growth percentages from 2020 to 2024.
  • Show increase/decrease using trend indicators.

- **Analysis Results:**
  • The highest YoY growth occurred in 2024.



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Dashboard Review: